TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time**: from 09/2026 To 03/2027

(*) Profession: Software Engineer **Specialty**: SE [x]

(*) Kinds of person make registers: Lecturer [x] Students

1. Register information for supervisor (if have)

No.	Full name	Phone	**E-Mail**	**Title**
Supervisor	Đặng Ngọc Minh Đức		ducdnm2@fpt.edu.vn	
	Đỗ Hữu Tiến			

2. Register information for students (if have)

No.	**Full name**	**Student code**	**Phone**	**E-mail**	**Role in Group**
1	Lê Thị Yến Vi	SE182544	0985868349	ViLTYSE182544@fpt.edu.vn	Leader
2	Trần Kim Nhã	SE182290	0933102278	NhaTKSE182290@fpt.edu.vn	Member
3	Nguyễn Quốc Bảo Huy	SE180211	0911649183	HuyNQBSE180211@fpt.edu.vn	Member
4	Trương Nguyễn Thái Bình	SE182370	0869371050	BinhTNTSE182370@fpt.edu.vn	Member

3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

3.1.1. English: ***Drone Delivery and Smart Locker Management System (DDSLMS)***

3.1.2. Vietnamese: ***Hệ thống quản lý giao nhận hàng bằng Drone và Smart Locker***

(*) 3.2. Main proposal content (including result and product)

a) Context:

The rapid growth of e-commerce has significantly increased the demand for fast, efficient, and contactless last-mile delivery services. Smart Locker systems have been widely deployed in residential areas, universities, and commercial centers, providing a practical infrastructure for parcel pickup and delivery. However, most existing drone delivery solutions focus primarily on flight control, routing algorithms, or simulation environments without offering an integrated

platform that combines delivery management, drone fleet management, maintenance management, real-time monitoring, and Smart Locker operations.

This project aims to develop an integrated Drone Delivery and Smart Locker Management System (DDSLMS) based on the Hub-to-Smart Locker delivery model. The system supports autonomous delivery operations, real-time monitoring, maintenance management, and AI-assisted operational support, providing a complete software solution for smart logistics applications.

b) Proposed Solutions:

The proposed system is a full-stack logistics platform integrating drone delivery services with a Smart Locker network. Drones operate between a central hub and Smart Locker stations, enabling secure and contactless parcel delivery.

The system provides the following core capabilities:

- Delivery order management and tracking.
- Drone fleet management and dispatching.
- Real-time drone telemetry simulation and monitoring.
- Smart Locker management and parcel collection using a PIN at the kiosk, or from the mobile app over Bluetooth or after scanning the locker QR code.
- Incident reporting and resolution management.
- Drone maintenance scheduling and repair management.
- AI-powered knowledge support using Retrieval-Augmented Generation (RAG).

c) Functional Requirements:

Module 1: Delivery Management

- Create, update, and manage delivery orders.
- Monitor order status throughout the delivery lifecycle.
- Estimate delivery time (ETA).
- Assign to each delivery an idle drone whose battery is above the admin-set minimum and that is not under maintenance.
- Track delivery history and performance statistics.
- Support real-time order tracking through WebSocket communication.

Module 2: Smart Locker Management

- Manage Smart Locker locations and compartments.
- Generate a PIN for parcel pickup; open the box from the mobile app over Bluetooth or after scanning the locker QR code.
- Support parcel deposit and retrieval workflows.
- Automatically release expired locker compartments.
- Monitor locker utilization and operational status.

Module 3: Incident Reporting and Management

- Submit locker compartment and delivery incident reports with photo evidence.
- Assign and track incident tickets with SLA deadlines for technicians.
- Hold affected locker compartments from new orders until inspection and resolution.
- Notify customers and technicians of incident status and resolution updates.

Module 4: Maintenance Management

- Schedule periodic drone maintenance.
- Record maintenance and repair logs.
- Exclude drones under maintenance from delivery assignment.
- Generate maintenance notifications and alerts.

Module 5: AI Knowledge Support

- Provide document-based question answering using RAG.
- Support retrieval of operational procedures and technical documentation.

d) Non-Functional Requirements:

Performance

- 95% of read API requests under 1 s and write requests under 2 s at 100 concurrent sessions.
- Order status and drone stage updates reach the app within 2 s (95th percentile).
- Box open result shown within 5 s in 95% of opens.

Reliability

- Monthly API uptime of at least 99%.
- Replayed payment webhooks are never counted twice.

Security

- JWT-based authentication.
- Role-Based Access Control (RBAC).
- Secure access to operational and maintenance data.
- A box is locked after 5 wrong PIN entries; Bluetooth open tokens are single-use.

Scalability

- Support deployment using Docker containers.
- Microservice-ready architecture.
- CI/CD integration through GitHub Actions.

e) Theory & Practical:

Backend

- Java Spring Boot
- RESTful APIs
- PostgreSQL
- JPA/Hibernate

Real-Time Communication

- WebSocket
- Socket.IO

Artificial Intelligence

- Retrieval-Augmented Generation (RAG)
- Large Language Model (LLM) APIs
- pgvector

Frontend

- ReactJS / NextJS
- Tailwind CSS
- Chart.js / Recharts

Mobile Application

- Flutter

DevOps

- Docker
- Docker Compose
- GitHub Actions

f) Products (Expected Deliverables):

Software Products

- Drone Delivery Management Web Application.
- Smart Locker Management System.
- Real-Time Monitoring Dashboard.
- AI Knowledge Support Module.
- Customer Mobile Application.

g) Proposed Tasks:

Phase 1: Requirement Analysis and System Design

- Literature review.
- Requirement analysis.
- Database and system architecture design.
- Software Requirement Specification development.

Phase 2: Core System Development

- Delivery Management Module.
- Drone Fleet Management Module.
- Smart Locker Management Module.
- Drone Simulator Service.

Phase 3: Advanced Features

- Real-time tracking and monitoring.
- Incident Reporting and Management Module.
- Maintenance Management Module.
- Notification services.

Phase 4: Deployment and Finalization

- Containerization and CI/CD deployment.
- Security and performance testing.
- AI Knowledge Support integration.
- Final report preparation.

3.3. Objectives, Scope & Methodology

a. Objectives:

General Objective

To develop and evaluate an integrated Drone Delivery and Smart Locker Management System that supports efficient and contactless last-mile delivery operations.

Specific Objectives

- Develop a comprehensive drone delivery management platform.
- Build a Drone Simulator Service for demonstration and testing.

- Integrate AI-powered knowledge support for operational assistance.

b. Scope & Methodology:

Scope

In Scope

- Delivery Management.
- Drone Fleet Management.
- Smart Locker Management.
- Drone Simulator Service.
- Incident Reporting and Management.
- Maintenance Management.
- AI Knowledge Support.

Out of Scope

- Physical drone control and flight operations.
- Computer vision-based object detection.
- Machine-learning-based route optimization.
- Predictive maintenance systems.

Methodology

- Literature Review.
- Requirement Analysis.
- System Design and Development.
- Agile Software Development.

4. Other comments (propose all relative things if have):

Version 2.0 (08/11/2026): research part removed (Battery-aware drone assignment, comparison of assignment strategies, experimental evaluation report, conference paper); AI summaries of operational reports removed; Incident Reporting and Management module added; non-functional requirements updated; parcel pickup methods stated as built (PIN at the kiosk; Bluetooth or locker QR scan in the app). Details: Record of Changes (BM03).

HCM, date 17/08/2026

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*

Registration Form – Record of Changes

1. Project Information

Project code	FA26SE002 (group GFA26SE32)
Project name	Drone Delivery and Smart Locker Management System (DDSLMS) – Hệ thống quản lý giao nhận hàng bằng Drone và Smart Locker (product name: Lock.R)
Team	Lê Thị Yến Vi (SE182544, leader) · Trương Nguyễn Thái Bình (SE182370) · Nguyễn Quốc Bảo Huy (SE180211) · Trần Kim Nhã (SE182290)
Supervisor	Đặng Ngọc Minh Đức · Đỗ Hữu Tiến
Current registration version	1.0 – registration form signed 17/08/2026
Proposed version	2.0
Version	1.0
Date	08/11/2026

2. Changes

☐ No changes to the current registration form

No.	Section	Current content	Proposed content	Reason	Proposed by
1	3.2 b) Proposed Solutions; 3.2 c) Module 1	Capability "Battery-aware drone assignment strategy"; FR "Implement Battery-aware Drone Assignment".	Capability removed. FR: "Assign to each delivery an idle drone whose battery is above the admin-set minimum and that is not under maintenance."	Findings from Iteration 1: the pilot has one real drone (DRONE-S550-07), so comparing drone assignment strategies gives no meaningful result. Effort moved to real hardware integration (cabinet CAB-TU01, drone camera, servo parcel release) and to a named technician being responsible for every flight (Design Brief D4). Only the research scope is reduced; the delivery workflow and its business rules are unchanged.	Team
2	3.2 f) Products; 3.2 g) Proposed Tasks	Research Outputs (experimental evaluation report, conference paper); Phase 3 "Experimental evaluation of drone assignment strategies"; Phase 4 "Conference paper preparation".	Removed. Phase 3 renamed "Advanced Features".	Same as No. 1.	Team

No.	Section	Current content	Proposed content	Reason	Proposed by
3	3.3 Research Information	Research question; research objectives on assignment strategies; methodology "Experimental Evaluation", "Comparative Performance Analysis"; expected scientific contribution.	Section renamed "Objectives, Scope & Methodology". Removed the research question, the assignment-strategy objectives, the two experimental methods and the scientific contribution. Kept the general objective, In/Out of Scope and the remaining methods; the simulator objective now reads "for demonstration and testing".	Same as No. 1.	Team
4	3.2 c) Module 4 AI Knowledge Support	Generate summaries of operational reports.	Removed. Document-based question answering with cited sources (RAG) is kept.	Not aligned with the project scope: the assistant answers operational questions from the knowledge base with cited sources, and operational figures are already shown on the admin dashboard and revenue reports (UC25).	Team
5	3.2 d) Non-Functional Requirements	p95 API response < 300 ms; real-time update < 2 s; telemetry every 3–5 s; availability > 95 % during testing; JWT; RBAC; secure access.	p95 read <= 1 s / write <= 2 s at 100 sessions; status and drone stage updates <= 2 s (p95); box open result <= 5 s in 95 % of opens; monthly uptime >= 99 %; replayed payment webhooks never counted twice; box locked after 5 wrong PINs and single-use Bluetooth tokens. JWT, RBAC, secure access and scalability kept. Same as Design Brief §7 N1–N6.	Constraint from technology: thresholds set to what can be measured on the deployed VM (2 vCPU, 8 GB RAM, SRS §4.2.3) and kept identical to Design Brief §7.	Team
6	3.2 b) Proposed Solutions; 3.2 c) Module 2	"Smart Locker management and parcel collection using PIN/QR authentication"; "Generate PIN and QR codes for parcel pickup".	"… parcel collection using a PIN at the kiosk, or from the mobile app over Bluetooth or after scanning the locker QR code"; "Generate a PIN for parcel pickup; open the box from the mobile app over Bluetooth or after scanning the locker QR code".	Findings from Iteration 1: the kiosk accepts a typed PIN only. QR is used the other way round: the app scans the QR sticker on the locker and opens the box over the Internet, or opens it over Bluetooth with a single-use token checked on the Pi (Design Brief D2). No device scans the order QR shown in the app.	Team
7	3.2 b) Proposed Solutions; 3.2 c) Module 3; 3.2 g) Phase 3; 3.3 b) In Scope	Modules 1–4 only (Delivery Management, Smart Locker Management, Maintenance Management, AI Knowledge Support); incident reporting not listed as a separate module.	Added "Module 3: Incident Reporting and Management" before Maintenance Management (renumbered to Module 4; AI Knowledge Support renumbered to Module 5), with matching items in Proposed Solutions, Phase 3 and In Scope.	Findings from Iteration 1: incident reporting and resolution (box incidents with SLA tickets and dropped-parcel recovery) is one of the four core workflows owned and demonstrated by the team (WF-03, Design Brief §8 and Appendix A1).	Team

3. Decision at Review 2 (completed by reviewers)

No.	Decision (Accepted / Adjusted / Rejected)	Reviewer note
1		
2		
3		
4		
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